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# Cross-View Distillation and Adaptive Masking for Incomplete Multi-View Multi-Label Classification

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## Abstract

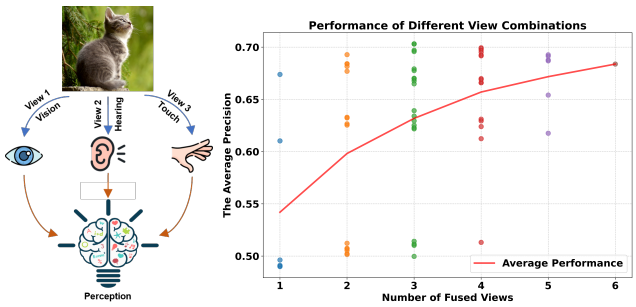
*While existing incomplete multi-view multi-label learning methods have achieved promising performance, few studies have focused on the issue of multi-view imbalance. Existing methods using gradient modulation or alternating optimization strategies alleviate this problem but often oversimplify the interaction between views, resulting in persistently poor performance. In response to the challenge, we propose the Cross-view Distillation and Adaptive Masking (CDAM) framework, a novel approach designed to achieve balanced multi-view optimization for the challenging double incomplete multi-view multi-label learning tasks. First, to overcome the performance bottleneck of views, we design a cross-view distillation module. This module aligns low-quality student representations with high-quality teacher representations, thereby effectively mitigating the multi-view imbalance problem. Second, recognizing that distillation may not rectify all low-quality views, we introduce a subsequent adaptive masking module to perform an explicit quality assessment. This module dynamically identifies and masks out any remaining unreliable representations before multi-view fusion, thus preventing low-quality information from corrupting the fused representation. Extensive comparisons with nine state-of-the-art methods on six datasets validate the effectiveness and stability of our method.*

## 1. Introduction

Real-world learning is inherently a multi-view process, wherein we comprehend the same phenomenon through various views such as visual observation, linguistic description, and physical interaction. As shown in Fig. 1a, the brain is capable of jointly embedding information from different senses, a process known in cognitive neuroscience as multi-sensory integration [9, 28]. This cross-view integration not only enhances our reasoning and understanding through synergistic representations across views but also establishes robust cognitive associations that allow us to perceive the world consistently even from partial sensory inputs [44]. Drawing inspiration from such insights, learning from multi-view data collected from various sensors has emerged as a research hotspot in the machine learning community, finding wide-ranging applications in fields such as medical image analysis [3, 23] and sentiment analysis [14]. However, in real-world scenarios, we frequently encounter the dual challenge of missing views (e.g., sensor malfunction) and missing labels (e.g., high annotation costs). To address this, Incomplete Multi-view Multi-label Learning (iM2L) aims to tackle the dual challenge of incompleteness, striving to effectively leverage multi-view information under partial supervision and incomplete data.

From an information-theoretic perspective, the advantage of multi-view learning lies in its capacity to capture richer and more comprehensive semantics. Since the model can access and fuse information from multiple data sources, the derived representations are significantly more robust and expressive than those of a single-view model [24, 40]. Therefore, models trained on multi-view data often outperform their counterparts trained on only a single view [36]. However, previous studies [10, 32] have also shown that significant heterogeneity often exists between different views, such as differences in data distribution, feature dimensionality, and noise levels. This inherent view disparity can lead to inconsistent learning difficulties and convergence speeds across views. Consequently, forcing the optimization of the entire multi-view model under a monolithic and static joint-training objective can lead to suboptimal performance [35]. As illustrated in Fig. 1b, we evaluate the classification performance of various view combinations on the complete Pascal07 dataset. The results reveal significant performance discrepancies across different combinations. The strategy

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(a) Brain perception. (b) View combination performance

Figure 1. Illustration of the multi-view learning paradigm and its performance advantages. (a) An analogy for multi-view perception. (b) Performance vs. number of fused views using a view-level weighted fusion strategy.

of combining a fixed set of dominant views, though optimal for complete data, is exceptionally fragile in practice. In real-world scenarios with missing views, the loss of just one critical view may cause a degradation in the model's overall performance, highlighting the pressing need for dynamic and robust fusion strategies.

Recently, several studies [38, 43, 47] have proposed methods to mitigate the view imbalance problem. By dropping representations from dominant views, curtailing their gradient flow in back-propagation, or constraining multi-view representations with pre-trained single-view encoders, these methods effectively force the model to enhance its learning from weaker views.

While such strategies successfully redirect attention to lagging views, they rely on a critical assumption: that weaker views are merely under-optimized. However, we argue that this assumption is flawed. As we observed, there is an intrinsic performance gap between different views, and some views suffer from an inherent representation ceiling due to limited information capacity or high noise, meaning that they cannot be improved solely by self-optimization. Consequently, forcing the model to learn from such ceiling-limited views impedes the optimization of other views, thereby resulting in a degradation of overall performance.

To address this limitation, we propose the Cross-view Distillation and Adaptive Masking (CDAM) framework. It is designed to alleviate the intertwined problems of imbalance and noise inherent in the iM2L task. First, we introduce a cross-view distillation strategy. This approach leverages the learned view-level fusion weights to identify the most dominant view for each sample, designating its representation as a reliable teacher. Subsequently, the representations of all other views are explicitly trained to align with the teacher's representations. Furthermore, for low-quality representations that remain resistant to knowledge distillation, we introduce an adaptive masking strategy. This mechanism acts as a crucial safeguard, identifying and masking out these unreliable representations before fusion to prevent the corruption of high-quality features. Finally, to address the concurrent challenge of label imbalance, our framework incorporates a gradient modulation strategy based on the label frequency distribution. This ensures that long-tail labels receive sufficient optimization. In summary, our contributions are as follows:

- We reveal an inherent performance ceiling in certain views: even when trained exhaustively, their performance remains limited and hard to surpass.
- We propose a framework designed to alleviate multi-view imbalance in iM2L. First, it fosters balanced training of multiple views via cross-view distillation. Second, it utilizes an adaptive masking mechanism to prevent quality degradation of the final fused representation.
- Extensive experimental results demonstrate our method's superior performance and strong robustness across a wide range of data missing scenarios.

## 2. Related Work

### 2.1. iM2L

Incomplete Multi-view Multi-label Learning aims to leverage multi-view information for downstream tasks under the dual challenges of missing views and partial label supervision. At the multi-view level, a dominant research paradigm is built upon the multi-view consistency assumption, which focuses on learning a consistent representation shared across all views [17, 45]. Furthermore, some studies [16, 33] attempt to leverage representation decoupling to exploit complementary information across views, thus addressing the challenge of missing data. Another line of work addresses missing views via imputation [18, 48], but the unpredictable quality of the generated data often introduces detrimental noise, leading to severe performance degradation. While these approaches have achieved promising results, the critical issue of multi-view imbalance remains largely unexplored [5, 22]. To address this, we design a cross-view distillation and adaptive masking strategy to mitigate the imbalance problem in iM2L tasks.

Regarding multi-label learning, a primary issue stems from the long-tail label distribution common in real-world data [21, 41], which creates an imbalance where the optimization is monopolized by high-frequency labels. Consequently, the gradients of the low-frequency labels are overwhelmed, leading to their ineffective learning by the model. To tackle this challenge, we introduce a gradient modulation strategy informed by the multi-label distribution. This approach down-weights the gradients originating from high-frequency head labels, allowing for sufficient optimization of the low-frequency tail labels.

### 2.2. Imbalanced Multi-view Learning

Recent studies have revealed a critical imbalance problem in multi-view learning. Due to the heterogeneous information contained within different views, models are prone to overfitting to dominant views and suffer from imbalanced optimization, which ultimately constrains their performance [26, 34]. To address this challenge, recent research has led to several advanced technical paradigms. Representative approaches primarily include: teacher view guidance [4], where uni-modal teachers are pre-trained on single views to guide the learning of the weaker view's encoder within the multi-view framework; gradient modulation [8, 13, 38, 43], which dynamically adjusts the back-propagated gradients for different views; and alternating view optimization [10, 37, 47], which involves alternately freezing and optimizing view-specific parameters during training. The core idea of these methods is to slow the rapid convergence of dominant views while simultaneously promoting the optimization of weaker views. However, these approaches often neglect cross-view information interactions. Moreover, existing strategies often assume complete data during training, making them fragile when views are missing randomly. This becomes especially problematic when a weaker view is inherently hard to optimize on its own. In this scenario, the crucial challenge is not just to boost the weaker view's training, but to effectively utilize information from stronger views to guide its optimization.

## 3. Methodology

In this section, we propose the Cross-view Distillation and Adaptive Masking framework (CDAM), as illustrated in Fig. 2. This framework mainly consists of feature extraction, multi-label classification, and two core modules: cross-view distillation and adaptive view masking.

### 3.1. Notation

We first introduce the notation and problem definition. Consider a multi-view multi-label dataset $\{\mathbf{X}^{(v)}\}_{v=1}^{m}$, where $\mathbf{X}^{(v)} \in \mathbb{R}^{n\times d_v}$, $n$ denotes the total number of samples, $d_v$ denotes the feature dimension of the $v$-th view, and $m$ represents the number of views. Correspondingly, the label matrix is defined as $\mathbf{Y} \in \{0,1\}^{n\times c}$, where $c$ is the number of labels. Furthermore, we introduce two indicator matrices to model the data incompleteness. We define a view mask matrix $\mathbf{M} \in \{0,1\}^{n\times m}$, where $M_{i,v} = 1$ indicates that the feature of the $v$-th view for the $i$-th sample is observed, and $M_{i,v} = 0$ indicates that it is missing. Similarly, we define a label mask matrix $\mathbf{G} \in \{0,1\}^{n\times c}$, where $G_{i,j} = 1$ indicates that the $j$-th label of the $i$-th sample is known, while $G_{i,j} = 0$ means it is unknown.

### 3.2. Multi-view Representation Learning

Given that the raw data from different views often have heterogeneous dimensionalities, we employ a set of view-specific encoders to facilitate effective cross-view interaction. Each encoder projects the features of its corresponding view into a shared latent space with a common dimension. This process is formally expressed as $\{\mathbf{Z}^{(v)} = \mathcal{E}^{(v)}(\mathbf{X}^{(v)})\}_{v=1}^{m}$, where $\mathbf{X}^{(v)}$ is the input data for the $v$-th view, $\mathcal{E}^{(v)}$ is the corresponding view-specific encoder, and $\mathbf{Z}^{(v)} \in \mathbb{R}^{n\times d_s}$ represents the latent representations for all samples from that view, with $d_s$ denotes the dimension of the shared latent space. To preserve information and prevent representation collapse, we also introduce a view-specific decoder for each view, which serves as a reconstruction regularizer by restoring the available original view data: $\{\mathcal{D}^{(v)}(\mathbf{Z}^{(v)}) = \hat{\mathbf{X}}^{(v)}\}_{v=1}^{m}$. The corresponding multi-view reconstruction loss is defined as follows:

$$\mathcal{L}_{rec} = \frac{1}{\sum_{i=1}^{n}\sum_{v=1}^{m} M_{i,v}} \sum_{i=1}^{n}\sum_{v=1}^{m} \|\mathbf{x}_i^{(v)} - \hat{\mathbf{x}}_i^{(v)}\|_2^2 M_{i,v}, \tag{1}$$

where $\mathbf{x}_i^{(v)}$ and $\hat{\mathbf{x}}_i^{(v)}$ denote the $v$-th view of the $i$-th original and reconstructed samples, respectively.

### 3.3. Cross-view Distillation

The performance gap between views, as observed in Fig. 1b, is due to their different levels of information purity. For example, in foggy weather, infrared images often contain more useful information than RGB images. To address the multi-view imbalance problem, inspired by [4, 46], we propose a novel Cross-view Distillation Module (CDM). Unlike existing methods that remain largely at the level of passively modulating the learning rates of each view via gradient modulation or alternating optimization, our CDM actively leverages the information interplay between views to resolve the multi-view imbalance problem. Specifically, at each training step, we first identify the most informative view among the observed views based on the learned global preference, and then utilize its high-quality representation to distill the other available student view representations.

In the multi-view representation fusion stage, we employ a fusion strategy guided by a learnable view-level vector before obtaining the final fused representation $\{\mathbf{z}_i\}_{i=1}^{n}$:

$$\mathbf{z}_i = \sum\nolimits_{v=1}^{m} w_i^{(v)} \mathbf{z}_i^{(v)}, \quad i = 1, ..., n \tag{2}$$

where $w_i^{(v)}$ is the fusion weight for the $v$-th view of the $i$-th sample that can be calculated by the following formula:

$$w_i^{(v)} = \frac{\exp(l_i^{(v)})}{\sum_{k=1}^{m}\exp(l_i^{(k)})}, \quad l_i^{(v)} = \begin{cases} q^{(v)}, & M_{i,v} = 1 \\ -\infty, & M_{i,v} = 0 \end{cases} \tag{3}$$

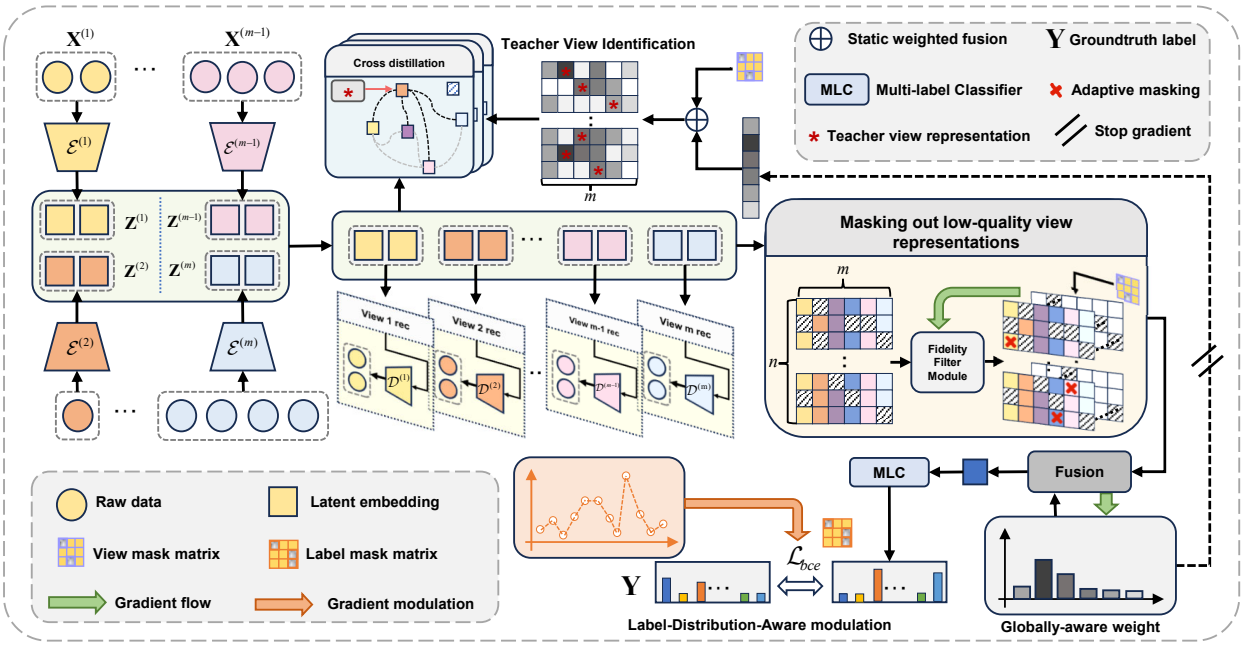



Figure 2. The framework of our proposed CDAM. The model consists of two key modules: The Cross-view Distillation Module (CDM) first identifies the teacher view based on fusion weights, and then compels weaker student views to align with its representation, thereby enhancing their quality. Subsequently, the Adaptive view Masking Module (AMM) acts as a filter, dynamically identifying and masking out any post-distillation representations that remain unreliable to ensure a robust final fusion. Finally, the classification loss is modulated by a label-distribution-aware mechanism to handle label imbalance.

where $\mathbf{q} \in \mathbb{R}^m$ is a global, learnable view preference vector, and $q^{(v)}$ denotes its $v$-th scalar element. Based on the fusion weights of the instance, we can identify the teacher view. Specifically, the teacher's view for the $i$-th sample is defined as the view with the highest fusion weight among all its observed views:

$$\mathbf{z}_i^* = \mathbf{z}_i^{(v_i^*)}, \quad v_i^* = \underset{v \in \{1,\dots,m\}}{\arg\max} \{w_i^{(v)} | M_{i,v} = 1\}. \tag{4}$$

Correspondingly, we define the set of student views for the $i$-th sample, denoted as $S_i$, as the set of all its observed views excluding the teacher view $\mathbf{z}_i^*$. Formally, this can be expressed as: $S_i = \{v \mid M_{i,v} = 1 \text{ and } v \neq v_i^*\}$. Finally, the cross-view distillation loss is calculated by aggregating the losses over all effective teacher-student pairs for each sample, as formulated below:

$$\mathcal{L}_{dis} = \frac{\sum_{i=1}^{n} \sum_{v \in S_i} \|\mathbf{z}_i^{(v)} - \text{sg}[\mathbf{z}_i^*]\|_2^2}{\sum_{i=1}^{n} |S_i|}, \tag{5}$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operation, and we ignore samples with $|S_i| = 0$. This ensures a unidirectional flow of knowledge from the teacher view to the student views by fixing the teacher as a stable learning target. Consequently, this prevents the model from collapsing and enables effective knowledge distillation.

### 3.4. Adaptive View Masking

While the CDM aims to comprehensively enhance the representation quality of weaker views, the distillation process may still fall short of the desired effect for certain sample views that contain extreme noise or minimal information. Furthermore, static view-level weighting ignores instance-level variations, allowing low-quality views to contaminate the fused representation, which in turn degrades the overall performance of the model. To address this issue, inspired by [27, 29], we introduce the Adaptive view Masking Module (AMM). AMM operates via an instance-aware gating mechanism that makes a binary decision for each view of a given sample: to include it in the fusion or to mask it entirely. This end-to-end trainable gate acts as a dynamic filter, ensuring that the final representation is constructed exclusively from the most task-relevant views. Departing from static weighting based on global statistics, our AMM introduces a dynamic MLP-based scoring network $f_{scorer}$. It generates a score for each view's representation on the fly:

$$e_i^{(v)} = f_{scorer}(\mathbf{z}_i^{(v)}) \tag{6}$$

where $e_i^{(v)}$ represents the evidence collected by the model supporting the proposition that the $v$-th view of the $i$-th sample should be selected. However, view selection is inherently a discrete binary decision. This discrete nature ren-

ders the decision process non-differentiable, which in turn obstructs the flow of gradients during backpropagation.

To facilitate end-to-end training, we employ the Gumbel-Softmax reparameterization trick [12] to handle the discrete nature of view selection. Specifically, for each view, we first generate a two-dimensional score vector $\mathbf{t}_i^{(v)} = [e_i^{(v)}, -e_i^{(v)}]$ from its evidence. We then draw samples $\mathbf{g}_i^{(v)}$ from the $\mathrm{Gumbel}(0,1)$ distribution, which can be calculated using the inverse CDF [25]: $g_{i,k}^{(v)} = -\log(-\log(u_{i,k}^{(v)}))$, where $u_{i,k}^{(v)} \sim \mathrm{Uniform}(0,1)$, and $g_{i,k}^{(v)}$ denotes the $k$-th element of the Gumbel noise vector. To make the sampling process differentiable, we compute a continuous relaxation using the softmax function with a temperature parameter $\tau$:

$$\tilde{\mathbf{s}}_i^{(v)} = \mathrm{softmax}\left((\mathbf{t}_i^{(v)} + \mathbf{g}_i^{(v)})/\tau\right). \tag{7}$$

During the forward pass, to ensure a strict discrete binary decision, we apply the Straight-Through Estimator (STE) [1] to obtain the hard one-hot vector $\mathbf{s}_i^{(v)}$. Meanwhile, during backpropagation, the gradients are approximated and routed through the continuous variable $\tilde{\mathbf{s}}_i^{(v)}$:

$$\mathbf{s}_i^{(v)} = \mathrm{OneHot}(\arg\max_{k\in\{1,2\}} \tilde{s}_{i,k}^{(v)}). \tag{8}$$

Finally, the first dimension of this one-hot vector serves as the binary selection mask for the corresponding view:

$$s_i^{(v)} = \mathbf{s}_{i,1}^{(v)}. \tag{9}$$

At inference time, the model operates in a deterministic mode to ensure consistent outputs: $s_i^{(v)} = \mathbb{I}(\sigma(e_i^{(v)}) > 0.5)$, where $\mathbb{I}(\cdot)$ is the indicator function and $\sigma$ denotes the sigmoid function. The final filtering mask is then produced by combining the learned selection mask with the prior view mask $M$ through element-wise multiplication.

$$\bar{M}_{i,v} = M_{i,v} \cdot s_i^{(v)} \tag{10}$$

As formulated in Eq. (11), we derive the updated fusion weights $\bar{w}$ for the multi-view representations:

$$\bar{w}_i^{(v)} = \frac{\exp(l_i^{(v)})}{\sum_{k=1}^{m}\exp(l_i^{(k)})}, \quad l_i^{(v)} = \begin{cases} q^{(v)}, & \bar{M}_{i,v} = 1 \\ -\infty, & \bar{M}_{i,v} = 0 \end{cases} \tag{11}$$

Subsequently, we calculate the filtered fused representation $\bar{\mathbf{z}}_i$ based on the fusion weights: $\bar{\mathbf{z}}_i = \sum_{v=1}^{m} \bar{w}_i^{(v)} \cdot \mathbf{z}_i^{(v)}$. Furthermore, to prevent the AMM from converging to a trivial solution (i.e., selecting all views), we introduce a targeted masking penalty. We impose a penalty on the selection masks assigned to these known missing views:

$$\mathcal{L}_{mask} = \frac{\sum_{i=1}^{n}\sum_{v=1}^{m} s_i^{(v)} \cdot \mathbb{I}(M_{i,v} = 0)}{\sum_{i=1}^{n}\sum_{v=1}^{m} \mathbb{I}(M_{i,v} = 0)} \tag{12}$$

**Algorithm 1** Training process of **CDAM**.

**Input:** Incomplete multi-view data $\{\mathbf{X}^{(v)}\}_{v=1}^{m}$, incomplete multi-label $\mathbf{Y} \in \{0,1\}^{n\times c}$, view mask matrix $\mathbf{M}$, label mask matrix $\mathbf{G}$.
**Output:** Trained parameters of model.
**Initialization:** Initialize the parameters of the model and set hyper-parameters ($\alpha$, $\beta$, and epochs $E$)
**for** $e = 1, \ldots, E$ **do**
  Extract multi-view embedding features $\{\mathbf{Z}^{(v)}\}_{v=1}^{m}$ by encoders $\{\mathcal{E}(\mathbf{X}^{(v)})\}_{v=1}^{m}$.
  Reconstruct data by decoders $\{\mathcal{D}(\mathbf{Z}^{(v)})\}_{v=1}^{m}$ and compute the reconstruction loss by Eq. (1).
  Identify the teacher view $\mathbf{z}^*$ by Eq. (4) and compute the distillation loss by Eq. (5).
  Generate the instance-level binary selection mask by Eq. (9) and Eq. (10).
  Obtain the fused representation $\mathbf{z}$ and the filtered representation $\bar{\mathbf{z}}$ using their respective weights.
  Obtain the corresponding predictions $\mathbf{p}$ and $\bar{\mathbf{p}}$ using the multi-label classifier.
  Compute the improvement loss by Eq. (14).
  Compute the total loss by Eq. (15) and update all trainable parameters.
**end for**

### 3.5. Multi-label Classification

After the two stages of enhancement by the CDM and filtering by the AMM, we generate two distinct fused multi-view representations for the $i$-th sample: the distilled representation $\mathbf{z}_i$, and the filtered representation $\bar{\mathbf{z}}_i$.

These representations are fed into a shared multi-label classifier $f_{cls}$ to yield their respective predictions: $\mathbf{p}_i = f_{cls}(\mathbf{z}_i)$ and $\bar{\mathbf{p}}_i = f_{cls}(\bar{\mathbf{z}}_i)$. For each prediction, we compute the Binary Cross-Entropy (BCE) loss against its corresponding ground-truth label:

$$\mathcal{L}_{cls} = \frac{1}{\sum_{i=1}^{n}\sum_{j=1}^{c} G_{i,j}} \sum_{i=1}^{n}\sum_{j=1}^{c} G_{i,j} \cdot h_j \cdot \mathcal{L}_{BCE}(p_{i,j}, Y_{i,j}) \tag{13}$$

where $\mathbf{p}_{i,j}$ and $\mathbf{Y}_{i,j}$ denote the prediction and the ground-truth label of the $j$-th class for the $i$-th sample, respectively. And $h_j = (1/f_j)^{\gamma}$ is the balance weight for label $j$ in the overall multi-label BCE loss, which is derived from its smoothed inverse frequency. $f_j$ is the frequency of the $j$-th label and $\gamma$ is used to smooth the inverse frequency, thereby preventing the over-weighting of rare labels. By substituting the predictions $\mathbf{p}$ and $\bar{\mathbf{p}}$ into Eq. (13), we obtain the corresponding classification losses, denoted as $\mathcal{L}_{cls}$ and $\bar{\mathcal{L}}_{cls}$, respectively. Our intuition is that the view filtering process should not degrade performance. We therefore enforce this by penalizing the model only when the classification loss

from the filtered representation exceeds that of the original. To this end, we introduce the following improvement loss:

$$\mathcal{L}_{imp} = \bar{\mathcal{L}}_{cls} + \max(0, \bar{\mathcal{L}}_{cls} - \text{sg}[\mathcal{L}_{cls}]) + \mathcal{L}_{mask} \quad (14)$$

By integrating all the above modules, we arrive at the final overall objective function:

$$\mathcal{L}_{all} = \alpha \cdot \mathcal{L}_{rec} + \beta \cdot \mathcal{L}_{dis} + \mathcal{L}_{imp} \quad (15)$$

where $\alpha$ and $\beta$ are hyperparameters that control the trade-off between the different optimization objectives.

# 4. Experiments

## 4.1. Datasets and Evaluation Metrics

### 4.1.1. Datasets

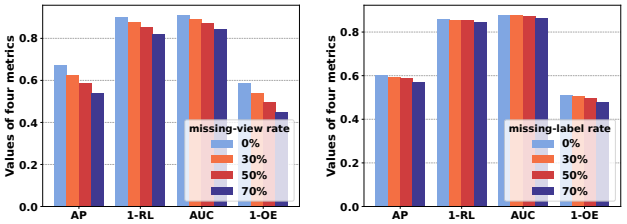
To comprehensively evaluate our proposed model, we conduct experiments on six benchmark multi-view multi-label datasets, following the settings of previous works [20, 30]. The selected datasets are: Corel5k [6], Pascal07 [7], Espgame [31], Iaprtc12 [**?** ], Mirflickr [11] and ODIR [2]. The first five datasets consist of six views: GIST, HSV, DenseHue, DenseSIFT, RGB, and LAB. In contrast, the ODIR dataset comprises five views: RGB, LAB, VGG, CLIP, and Wavelet. Moreover, we simulate missing data scenarios by randomly masking both views and labels. For instance, under a 50% missing rate setting, we mask 50% of the samples from each view, ensuring that each sample keeps at least one view, and for each specific label, we randomly mask 50% of its associated positive and negative samples to generate the incomplete multi-label data. The data is then divided into training (70%), validation (15%), and testing (15%) sets.

### 4.1.2. Evaluation Metrics

Following prior studies in iM2L [17, 42], we evaluate our method using six standard multi-label classification metrics: Average Precision (AP), Hamming Loss (HL), Ranking Loss (RL), Adapted Area Under Curve (AUC), One-Error (OE), and Coverage (Cov). For AP and AUC, a higher value indicates better performance. In contrast, for HL, RL, OE, and Cov, a lower value is better. To unify the evaluation, we report the latter four metrics as 1-HL, 1-RL, 1-OE, and 1-Cov. Consequently, for all metrics, a higher score signifies better performance.

## 4.2. Compared Methods

To validate the effectiveness of our method, we compare it with nine state-of-the-art methods, namely, iMVWL [30], NAIM3L [15], DIMC [39], DICNet [17], MTD [16], VCMN [5], AIMNet [18], MSLPP [22], and SIP [19]. All nine baseline methods are capable of natively handling incomplete multi-view multi-label classification. To ensure a fair comparison, we directly use the recommended hyperparameters provided in their official implementations. Some of these baselines have been discussed in Sec. 2.1. For a more detailed description of all comparison methods, please refer to the Appendix.




(a) various missing view rates. (b) various missing label rates.

Figure 3. Performance of four metrics on the Pascal07 dataset under (a) varying missing-view rates with a 50% missing-label rate, and (b) varying missing-label rates with a 50% missing-view rate.

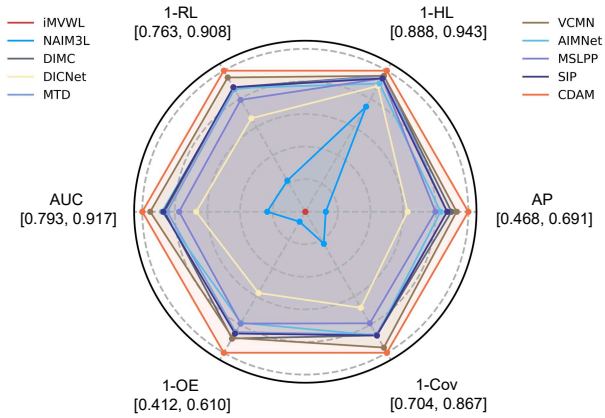



Figure 4. Visualization of the multi-metric performance comparison between our method (CDAM) and various baselines. For each metric, the value range is normalized, with the outer circle representing the optimal score.

## 4.3. Experimental Results

We conducted experiments in a setting with view and label missing rates of 50%. The results averaged over 10 independent runs are presented in Tab. 1, where our method is compared against nine state-of-the-art baselines across six datasets, and the best results for each metric are highlighted in gray. As observed from the experimental results, our CDAM consistently achieves state-of-the-art performance across all six datasets of varying scales.

A closer analysis of key baselines reveals the source of our method's superiority. For instance, while DICNet captures inter-view interactions, its contrastive learning strategy does not account for the inherent disparity in information quality across views. This oversight can lead to performance degradation when handling imbalanced multi-view

Table 1. Experimental results of ten methods on six datasets with 50% missing-view rate and 50% missing-label rate (the bottom right digit is the standard deviation).

| Data | Metric | iMVWL | NAIM3L | DIMC | DICNet | MTD | VCMN | AIMNet | MSLPP | SIP | **CDAM** |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Corel5k | AP | $0.283_{0.008}$ | $0.309_{0.004}$ | $0.353_{0.006}$ | $0.381_{0.004}$ | $0.415_{0.009}$ | $0.402_{0.011}$ | $0.400_{0.010}$ | $0.413_{0.008}$ | $0.418_{0.009}$ | $0.428_{0.006}$ |
| | 1-HL | $0.978_{0.000}$ | $0.987_{0.000}$ | $0.987_{0.000}$ | $0.988_{0.000}$ | $0.988_{0.000}$ | $0.987_{0.000}$ | $0.988_{0.000}$ | $0.988_{0.000}$ | $0.988_{0.000}$ | $0.988_{0.0002}$ |
| | 1-RL | $0.865_{0.005}$ | $0.878_{0.002}$ | $0.867_{0.001}$ | $0.882_{0.004}$ | $0.893_{0.004}$ | $0.900_{0.004}$ | $0.902_{0.003}$ | $0.901_{0.003}$ | $0.911_{0.003}$ | $0.915_{0.002}$ |
| | AUC | $0.868_{0.005}$ | $0.881_{0.002}$ | $0.870_{0.001}$ | $0.884_{0.004}$ | $0.896_{0.003}$ | $0.902_{0.004}$ | $0.905_{0.003}$ | $0.903_{0.004}$ | $0.913_{0.003}$ | $0.918_{0.002}$ |
| | 1-OE | $0.311_{0.015}$ | $0.350_{0.009}$ | $0.430_{0.013}$ | $0.468_{0.007}$ | $0.491_{0.012}$ | $0.478_{0.016}$ | $0.475_{0.018}$ | $0.485_{0.010}$ | $0.489_{0.016}$ | $0.498_{0.017}$ |
| | 1-Cov | $0.702_{0.008}$ | $0.725_{0.005}$ | $0.690_{0.003}$ | $0.727_{0.011}$ | $0.749_{0.009}$ | $0.770_{0.009}$ | $0.771_{0.005}$ | $0.766_{0.009}$ | $0.787_{0.009}$ | $0.801_{0.006}$ |
| Pascal07 | AP | $0.437_{0.018}$ | $0.488_{0.003}$ | $0.532_{0.002}$ | $0.505_{0.012}$ | $0.551_{0.004}$ | $0.560_{0.010}$ | $0.548_{0.008}$ | $0.544_{0.010}$ | $0.555_{0.010}$ | $0.588_{0.007}$ |
| | 1-HL | $0.882_{0.004}$ | $0.928_{0.001}$ | $0.931_{0.001}$ | $0.929_{0.001}$ | $0.932_{0.001}$ | $0.932_{0.001}$ | $0.931_{0.001}$ | $0.932_{0.001}$ | $0.931_{0.001}$ | $0.935_{0.001}$ |
| | 1-RL | $0.736_{0.015}$ | $0.783_{0.001}$ | $0.813_{0.000}$ | $0.783_{0.008}$ | $0.831_{0.003}$ | $0.837_{0.005}$ | $0.831_{0.004}$ | $0.819_{0.006}$ | $0.830_{0.004}$ | $0.855_{0.004}$ |
| | AUC | $0.767_{0.015}$ | $0.811_{0.001}$ | $0.833_{0.002}$ | $0.809_{0.006}$ | $0.851_{0.003}$ | $0.857_{0.005}$ | $0.851_{0.004}$ | $0.841_{0.004}$ | $0.850_{0.005}$ | $0.873_{0.004}$ |
| | 1-OE | $0.362_{0.023}$ | $0.421_{0.006}$ | $0.452_{0.006}$ | $0.427_{0.015}$ | $0.459_{0.007}$ | $0.467_{0.015}$ | $0.461_{0.013}$ | $0.466_{0.014}$ | $0.464_{0.018}$ | $0.499_{0.010}$ |
| | 1-Cov | $0.677_{0.015}$ | $0.727_{0.002}$ | $0.760_{0.001}$ | $0.731_{0.006}$ | $0.784_{0.003}$ | $0.791_{0.005}$ | $0.783_{0.004}$ | $0.771_{0.003}$ | $0.783_{0.006}$ | $0.810_{0.005}$ |
| Espgame | AP | $0.244_{0.005}$ | $0.246_{0.002}$ | $0.287_{0.002}$ | $0.297_{0.002}$ | $0.306_{0.003}$ | $0.308_{0.003}$ | $0.305_{0.004}$ | $0.310_{0.004}$ | $0.311_{0.004}$ | $0.311_{0.006}$ |
| | 1-HL | $0.972_{0.000}$ | $0.983_{0.000}$ | $0.983_{0.000}$ | $0.983_{0.000}$ | $0.983_{0.000}$ | $0.983_{0.000}$ | $0.983_{0.000}$ | $0.983_{0.000}$ | $0.983_{0.000}$ | $0.983_{0.000}$ |
| | 1-RL | $0.808_{0.002}$ | $0.818_{0.002}$ | $0.821_{0.000}$ | $0.832_{0.001}$ | $0.837_{0.002}$ | $0.843_{0.002}$ | $0.846_{0.002}$ | $0.843_{0.002}$ | $0.849_{0.002}$ | $0.852_{0.003}$ |
| | AUC | $0.813_{0.002}$ | $0.824_{0.002}$ | $0.826_{0.000}$ | $0.836_{0.001}$ | $0.842_{0.002}$ | $0.847_{0.002}$ | $0.850_{0.002}$ | $0.847_{0.002}$ | $0.853_{0.002}$ | $0.856_{0.003}$ |
| | 1-OE | $0.343_{0.013}$ | $0.339_{0.003}$ | $0.430_{0.004}$ | $0.439_{0.007}$ | $0.447_{0.009}$ | $0.454_{0.011}$ | $0.442_{0.011}$ | $0.457_{0.012}$ | $0.455_{0.007}$ | $0.446_{0.013}$ |
| | 1-Cov | $0.548_{0.004}$ | $0.571_{0.003}$ | $0.564_{0.003}$ | $0.593_{0.003}$ | $0.602_{0.004}$ | $0.618_{0.006}$ | $0.624_{0.005}$ | $0.622_{0.005}$ | $0.628_{0.005}$ | $0.642_{0.006}$ |
| Iaprtc12 | AP | $0.237_{0.003}$ | $0.261_{0.001}$ | $0.308_{0.001}$ | $0.323_{0.001}$ | $0.332_{0.003}$ | $0.331_{0.004}$ | $0.329_{0.005}$ | $0.340_{0.005}$ | $0.331_{0.006}$ | $0.352_{0.004}$ |
| | 1-HL | $0.969_{0.000}$ | $0.980_{0.000}$ | $0.980_{0.000}$ | $0.981_{0.000}$ | $0.981_{0.000}$ | $0.981_{0.000}$ | $0.981_{0.000}$ | $0.981_{0.000}$ | $0.980_{0.000}$ | $0.981_{0.000}$ |
| | 1-RL | $0.833_{0.002}$ | $0.848_{0.001}$ | $0.864_{0.000}$ | $0.873_{0.001}$ | $0.875_{0.002}$ | $0.881_{0.002}$ | $0.883_{0.003}$ | $0.882_{0.002}$ | $0.885_{0.003}$ | $0.895_{0.002}$ |
| | AUC | $0.835_{0.001}$ | $0.850_{0.001}$ | $0.864_{0.000}$ | $0.874_{0.000}$ | $0.876_{0.001}$ | $0.883_{0.002}$ | $0.885_{0.003}$ | $0.883_{0.002}$ | $0.886_{0.002}$ | $0.896_{0.002}$ |
| | 1-OE | $0.352_{0.008}$ | $0.390_{0.005}$ | $0.436_{0.001}$ | $0.468_{0.002}$ | $0.467_{0.005}$ | $0.465_{0.006}$ | $0.459_{0.080}$ | $0.474_{0.008}$ | $0.463_{0.009}$ | $0.489_{0.007}$ |
| | 1-Cov | $0.564_{0.005}$ | $0.592_{0.004}$ | $0.614_{0.001}$ | $0.649_{0.001}$ | $0.649_{0.004}$ | $0.673_{0.004}$ | $0.673_{0.006}$ | $0.672_{0.006}$ | $0.675_{0.007}$ | $0.700_{0.005}$ |
| Mirflickr | AP | $0.490_{0.012}$ | $0.551_{0.002}$ | $0.602_{0.002}$ | $0.589_{0.005}$ | $0.607_{0.004}$ | $0.611_{0.007}$ | $0.602_{0.004}$ | $0.615_{0.004}$ | $0.614_{0.004}$ | $0.631_{0.003}$ |
| | 1-HL | $0.839_{0.002}$ | $0.882_{0.001}$ | $0.888_{0.000}$ | $0.888_{0.002}$ | $0.891_{0.001}$ | $0.891_{0.002}$ | $0.890_{0.001}$ | $0.892_{0.001}$ | $0.891_{0.001}$ | $0.894_{0.001}$ |
| | 1-RL | $0.803_{0.008}$ | $0.844_{0.001}$ | $0.865_{0.001}$ | $0.863_{0.004}$ | $0.875_{0.001}$ | $0.876_{0.002}$ | $0.873_{0.002}$ | $0.879_{0.002}$ | $0.877_{0.002}$ | $0.886_{0.002}$ |
| | AUC | $0.787_{0.012}$ | $0.837_{0.001}$ | $0.852_{0.001}$ | $0.849_{0.004}$ | $0.862_{0.002}$ | $0.864_{0.001}$ | $0.861_{0.001}$ | $0.865_{0.002}$ | $0.860_{0.003}$ | $0.873_{0.002}$ |
| | 1-OE | $0.511_{0.022}$ | $0.585_{0.003}$ | $0.684_{0.001}$ | $0.637_{0.007}$ | $0.655_{0.004}$ | $0.660_{0.009}$ | $0.651_{0.006}$ | $0.667_{0.007}$ | $0.662_{0.008}$ | $0.683_{0.006}$ |
| | 1-Cov | $0.572_{0.013}$ | $0.631_{0.002}$ | $0.653_{0.002}$ | $0.652_{0.007}$ | $0.676_{0.004}$ | $0.680_{0.005}$ | $0.671_{0.004}$ | $0.679_{0.003}$ | $0.678_{0.003}$ | $0.694_{0.003}$ |
| ODIR | AP | $0.614_{0.007}$ | $0.594_{0.007}$ | $0.671_{0.007}$ | $0.661_{0.010}$ | $0.680_{0.006}$ | $0.683_{0.007}$ | $0.674_{0.010}$ | $0.672_{0.013}$ | $0.682_{0.009}$ | $0.701_{0.007}$ |
| | 1-HL | $0.781_{0.005}$ | $0.813_{0.008}$ | $0.875_{0.003}$ | $0.874_{0.002}$ | $0.876_{0.003}$ | $0.876_{0.003}$ | $0.873_{0.005}$ | $0.874_{0.003}$ | $0.876_{0.004}$ | $0.882_{0.002}$ |
| | 1-RL | $0.789_{0.005}$ | $0.751_{0.013}$ | $0.850_{0.005}$ | $0.839_{0.009}$ | $0.855_{0.003}$ | $0.857_{0.005}$ | $0.851_{0.006}$ | $0.847_{0.008}$ | $0.854_{0.006}$ | $0.870_{0.004}$ |
| | AUC | $0.839_{0.005}$ | $0.797_{0.014}$ | $0.889_{0.005}$ | $0.878_{0.008}$ | $0.894_{0.003}$ | $0.895_{0.004}$ | $0.891_{0.004}$ | $0.886_{0.006}$ | $0.892_{0.005}$ | $0.906_{0.003}$ |
| | 1-OE | $0.395_{0.010}$ | $0.389_{0.007}$ | $0.477_{0.013}$ | $0.466_{0.014}$ | $0.490_{0.010}$ | $0.495_{0.012}$ | $0.483_{0.016}$ | $0.481_{0.018}$ | $0.494_{0.018}$ | $0.518_{0.011}$ |
| | 1-Cov | $0.803_{0.005}$ | $0.776_{0.012}$ | $0.859_{0.005}$ | $0.849_{0.008}$ | $0.864_{0.003}$ | $0.865_{0.004}$ | $0.860_{0.006}$ | $0.857_{0.008}$ | $0.863_{0.005}$ | $0.877_{0.004}$ |

samples. Similarly, SIP leverages the Information Bottleneck principle to suppress non-shared information in pursuit of a consistent representation. However, the quality of this shared representation can be constrained by low-quality views, potentially hindering its effectiveness on samples that contain such inferior information. Additionally, we evaluate the model on the Pascal07 dataset under a wider range of view and label missing rates. The results are presented in Fig. 3. As observed, the model's performance exhibits a graceful degradation as the missing rate increases. Moreover, the model appears to be more sensitive to missing views than to missing labels. Furthermore, to comprehensively assess the model's robustness, in Tab. 2 we present a performance comparison between CDAM and several competing methods on the Corel5k, Pascal07, and Mirflickr datasets, under a setting with a 30% dual missing rate for both views and labels. The experimental results show that our method can deliver superior performance even when the missing rate is relatively low. Finally, we also evaluated its performance in the fully-observed setting, with the results illustrated in Fig. 4. The experimental results demonstrate that our CDAM model continues to perform exceptionally well in complete datasets. Performance results on the other datasets are provided in the Appendix.

### 4.4. Ablation Study

To validate the effectiveness of each key component in our proposed method, we conduct detailed ablation studies quantifying the contribution of each module. Specifically, we decompose our overall objective function into three core components: the multi-view representation learning and reconstruction loss $\mathcal{L}_{rec}$, the cross-view distillation loss $\mathcal{L}_{dis}$,

Table 2. Performance comparison with state-of-the-art methods on three datasets under a 30% dual missing rate for both views and labels. The best results for each metric are highlighted with a gray background. Our CDAM consistently outperforms all competing methods across all metrics and datasets.

| Dataset | Corel5k | | Pascal07 | | Mirflickr | |
|---|---|---|---|---|---|---|
| Method | AP | AUC | AP | AUC | AP | AUC |
| DIMC | 0.435 | 0.902 | 0.602 | 0.872 | 0.636 | 0.871 |
| DICNet | 0.441 | 0.901 | 0.570 | 0.851 | 0.625 | 0.866 |
| MTD | 0.479 | 0.917 | 0.607 | 0.879 | 0.643 | 0.875 |
| VCMN | 0.464 | 0.923 | 0.610 | 0.882 | 0.651 | 0.881 |
| AIMNet | 0.468 | 0.925 | 0.595 | 0.876 | 0.631 | 0.874 |
| MSLPP | 0.452 | 0.918 | 0.594 | 0.869 | 0.651 | 0.877 |
| SIP | 0.473 | 0.928 | 0.599 | 0.874 | 0.643 | 0.876 |
| CDAM | 0.488 | 0.935 | 0.637 | 0.895 | 0.664 | 0.887 |

Table 3. The ablation experimental results on the Corel5k and Pascal07 datasets, and all datasets are with 50% missing views, 50% missing labels and 70% training samples.

| Backbone | $\mathcal{L}_{rec}$ | $\mathcal{L}_{dis}$ | $\mathcal{L}_{imp}$ | Corel5k | | Pascal07 | |
|---|---|---|---|---|---|---|---|
| | | | | AP | AUC | AP | AUC |
| ✓ | | | | 0.3730 | 0.8748 | 0.5363 | 0.8362 |
| ✓ | ✓ | | | 0.3941 | 0.9061 | 0.5651 | 0.8564 |
| ✓ | | ✓ | | 0.3724 | 0.8979 | 0.5442 | 0.8545 |
| ✓ | | | ✓ | 0.3854 | 0.8912 | 0.5460 | 0.8441 |
| ✓ | ✓ | ✓ | | 0.4230 | 0.9154 | 0.5849 | 0.8707 |
| ✓ | ✓ | | ✓ | 0.4188 | 0.9155 | 0.5750 | 0.8655 |
| ✓ | | ✓ | ✓ | 0.3799 | 0.9029 | 0.5292 | 0.8467 |
| ✓ | ✓ | ✓ | ✓ | 0.4281 | 0.9178 | 0.5876 | 0.8733 |

and the adaptive view masking loss $\mathcal{L}_{imp}$. The results are presented in Tab. 3. To be clear, when we remove the Adaptive Masking Module corresponding to $\mathcal{L}_{imp}$, we still retain its component $\mathcal{L}_{cls}$ to provide the supervision signal for the classification task. The results show that all components are essential. The reconstruction loss acts as the foundation. Without a solid foundation in representation learning, neither the cross-view distillation nor the adaptive masking module can work effectively. Building on this base, introducing either cross-view distillation or adaptive masking alone increases performance, proving that each module is independently beneficial. The complete CDAM framework yields the highest overall performance, confirming that the two modules are highly complementary.

### 4.5. Parameter Sensitivity

To evaluate the sensitivity of the model to hyperparameters $\alpha$ and $\beta$, we performed a grid search on the Pascal07 dataset, with the results illustrated in Fig. 5. The figure

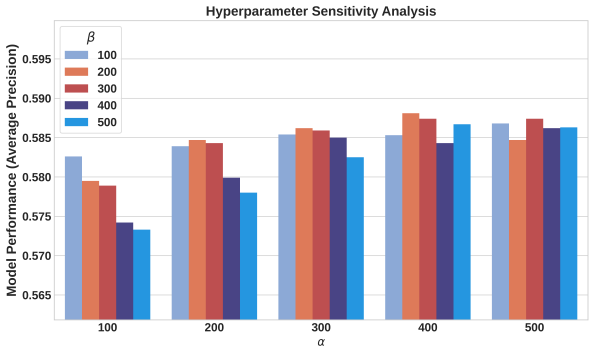



Figure 5. Sensitivity analysis of hyperparameters $\alpha$ and $\beta$ on the Pascal07 dataset. The model's performance (AP) is evaluated across a grid of $\alpha$ and $\beta$ values.

shows that as the value of $\alpha$ increases, the model's performance generally trends upward before stabilizing. However, the impact of $\beta$ exhibits a significant interaction with $\alpha$, and its optimal value depends on the choice of $\alpha$. For example, when $\alpha$ is small, average precision gradually decreases as $\beta$ increases; conversely, when $\alpha$ is relatively large, performance initially improves before dropping as $\beta$ increases. The peak of the precision exists within a relatively flat region where multiple parameter combinations achieve near-optimal results. Results on the remaining datasets are detailed in the Appendix. These findings confirm that our model consistently achieves strong performance across a wide range of parameter settings, demonstrating its robustness to hyperparameter selection.

## 5. Conclusion

This paper addresses the multi-view imbalance problem in incomplete multi-view multi-label learning (iM2L). Although existing gradient modulation and alternating optimization methods achieve strong results, they overlook a fundamental limitation: some views suffer from an inherent performance bottleneck. Consequently, multi-view imbalance persists even after exhaustive view-specific optimization. To address this, we propose the Cross-view Distillation and Adaptive Masking (CDAM) framework. First, CDAM leverages the cross-view distillation module to align weaker student view representations with reliable teacher view representations. Subsequently, an adaptive masking module dynamically filters out any remaining low-quality representations, preventing them from corrupting the final fusion. Experimental results on six datasets demonstrate that CDAM consistently outperforms state-of-the-art methods across various dual-missing scenarios. Ablation experiments further validate the effectiveness of each component.

## 6. Acknowledgements

This work is suppoted in part by Shenzhen Science and Technology Program (No. JCYJ20240813105135047), the Project of Xiangjiang Laboratory (25XJ03021), National Natural Science Foundation of China (No. 62372136), and Guangdong Basic and Applied Basic Research Foundation (No. 2024A1515030213).

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